

Phased Breakdown for ATS Project

Phase 1 – Foundation & Setup (2–3 weeks)

Goal: Lay the technical groundwork and define core system structure.

Tasks:

1. Project Planning & Documentation

- Finalize feature scope (must-have vs nice-to-have).
- Define data models (Users, Jobs, Applications, Evaluations).
- Draft user roles & permissions (Admin, Hiring Manager, Candidate).

2. Tech Stack Setup

- Setup GitHub repo & CI/CD pipeline.
- Setup Firebase (Auth, Firestore, Storage).
- Setup React environment & routing (React Router).
- Implement basic UI framework (Bootstrap/Tailwind).

3. Authentication

- Candidate sign-up/login.
- Hiring Manager & Admin login.
- Role-based access control.

Phase 2 – Job Requisition & Advertisement (4 weeks)

Goal: Build workflows for job creation & posting.

Tasks:

1. Job Requisition

- Create “New Job Requisition” form (fields: title, grade, department, JD upload).
- Approval workflow (single/multi-level).
- Auto-generate job reference number.
- Internal vs external advert flag.

2. Job Advertisement

- Standardized advert template.
- Enforce required fields (title, grade, location).
- Job status workflow (Pending, Active, Closed, Re-advertised).
- Auto-close jobs at deadline.
- Confidential/internal posting option.
- Optional RSS/social feed integration (defer if time is tight).

Phase 3 – Applications & Candidate Profiles (4–6 weeks)

Goal: Enable applicants to apply and track their submissions.

Tasks:

1. Candidate Profile

- Editable candidate dashboard (personal info, CV upload, cover letter).
- Prevent duplicate applications.
- Checklist for required docs.

2. Application Workflow

- Apply to job → auto-generate confirmation email.
- Prevent applying to closed jobs.
- Track applicant history (applied jobs, status).
- Detect internal/ex-employee (basic flag).
- Job alerts/subscriptions (optional if time permits).

3. Data Validation

- Pre-screening questions with scoring logic.
- Validate required fields before submission.

Phase 4 – Screening, Shortlisting & Interviews (6 weeks)

Goal: Build the recruitment engine.

Tasks:

1. Long Listing

- Compare applicant data vs job requirements.
- Scoring mechanism for pre-screening.
- Bulk status updates.
- Download CVs with auto-renaming.
- Candidate database search & filter.

2. Shortlisting

- Hiring manager dashboard to review longlist.

- Collaboration panel for notes & comments.

3. Interviews

- Interview scheduling & invite.
- Upload questions/templates.
- Record panel scores + comments.
- Auto-generate interview report.

Phase 5 – Reference Checks, Offers & Onboarding (4 weeks)

Goal: Complete recruitment cycle.

Tasks:

1. Reference Checks

- Auto-generate referee emails.
- Record responses.

2. Offers & Regrets

- Offer management (accept/reject status).
- Notify hiring manager.
- Send regret letters.
- Move hired candidate to onboarding stage.

3. Onboarding Handoff

- Export candidate details to onboarding system (Excel/JSON/other).
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Phase 6 – Reports & Analytics (3–4 weeks)

Goal: Build dashboards & reporting tools.

Tasks:

1. Standard Reports

- Time-to-hire, cost-per-hire, turnover, source of hire.
- Gender/age/nationality ratios.
- Vacancy cost & recruitment per department.

2. Visualizations

- Charts: Gantt, Pie, Bar (Recharts.js or Chart.js).

3. Ad-hoc Reports

- Search/filter-based exportable reports. (optional)
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Phase 7 – Security, Compliance & Maintenance (Ongoing)

Goal: Harden the system for real-world use.

Tasks:

1. Security

- Audit trail for user actions.
- No applicant editing after submission. **
- Strict role-based permissions.

2. Backup & Recovery

- Automated Firestore backups.

- Document recovery strategy.

3. Compliance

- GDPR/EEO statements in applications.
- Candidate consent checkbox.

4. Performance

- Optimize queries & indexes.
- Ensure mobile responsiveness.

1. Frontend (Client-Side)

You want a responsive, modern, and fast interface for **candidates, recruiters, and admins**.

● Frameworks & Libraries

- **React.js** (core UI framework)
- **TypeScript** (type safety & maintainability)
- **Vite** (fast build tool, better than CRA)
- **Bootstrap 5** (quick UI components) OR **Tailwind CSS** (if you want flexibility in design)
- **Redux Toolkit / React Query** (state management and API calls)
- **React Router** (routing between job listings, candidate dashboards, etc.)

● UI Enhancements

- **shadcn/ui** or **MUI (Material UI)** (for modals, tables, buttons, dropdowns, tabs, etc.)

- **Recharts / Chart.js** (analytics dashboards for recruiters)
 - **React Hook Form** (form handling for job applications & candidate profile forms)
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2. Backend (Server-Side)

The backend needs to handle **authentication, job postings, candidate data, resume parsing, analytics, and automation.**

- **Languages & Framework**
 - **Node.js + Express.js** (lightweight, scalable, widely supported)
 - **TypeScript** (to enforce type safety backend too)
- **Database**
 - **PostgreSQL** (structured data: jobs, candidates, recruiters, applications)
 - **MongoDB (optional)** (if you want flexibility for resume parsing data storage or logs)
- **Authentication**
 - **Firebase Auth** (simplifies login with Google/Email/Password, scalable) OR
 - **Auth0** (enterprise-ready if you need more SSO features)
- **File Handling**
 - **Firebase Storage / AWS S3 / Google Cloud Storage** (for resumes, offer letters, contracts, etc.)
- **Resume Parsing**
 - **Open-source NLP libraries:**
 - **PyMuPDF / pdfplumber** (extract text from resumes)
 - **spaCy / NLTK** (parse names, skills, experience)

- Optionally integrate **RChilli / Sovren (paid APIs)** if clients demand advanced parsing

3. Infrastructure & DevOps

- **Hosting**
 - **Frontend:** Vercel or Netlify (easy deploy for React/Vite)
 - **Backend:** Render / Railway / Heroku (free tiers for MVP) OR AWS/GCP/Azure for scale
- **Database Hosting**
 - **Firebase/ Supabase / Railway** (Postgres hosting with free tier)
 - **MongoDB Atlas** (if needed)
- **CI/CD**
 - **GitHub Actions** (automated testing & deployment on push) **
- **APIs**
 - **Nominatim (OpenStreetMap)** → for job location fields (free geocoding)
 - **Email API** → SendGrid / Mailgun (to send interview invites, rejections, etc.)
 - **Notifications** → Firebase Cloud Messaging (push alerts for recruiters & candidates)



4. Core Features Mapped to Tech

- **Job Posting**
 - Backend: Express.js + PostgreSQL

- Admin Panel UI: React + Bootstrap/Tailwind
 - **Resume Parsing**
 - Python microservice (Flask/FastAPI) for NLP resume parsing (connected to Node backend)
 - **Candidate Screening**
 - Rules stored in DB → scoring logic in backend
 - UI for recruiters to filter via React Data Tables
 - **Automation**
 - Background jobs using **BullMQ (Redis queue)** for tasks like auto-emailing candidates
 - **Dashboards**
 - Recruiter: React + Recharts (candidate pipeline, time-to-hire)
 - Candidate: Application status tracking
 - **Collaboration Tools**
 - Internal recruiter notes stored in DB
 - Shared feedback form → tied to candidate profiles
 - **Analytics & Reporting**
 - Backend aggregates → APIs → Charts in frontend
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5. Security & Compliance

- **OWASP Top 10 compliance** (prevent XSS, CSRF, SQL injection)
- **HTTPS** everywhere (SSL via Cloudflare or hosting provider)

- **GDPR readiness:** candidate consent, right to delete data
- **Database encryption:** passwords hashed (bcrypt), sensitive data encrypted at rest

6. Optional Enhancements

- **AI Matching** → Use **OpenAI embeddings** or **Haystack** to recommend best candidates for a job posting
- **Video Interviews** → Integrate **Twilio / Daily.co** for video interview scheduling
- **Calendars** → Google Calendar API / Microsoft Outlook API for interview scheduling sync
- **Chat Integration** → Firebase Realtime DB or WebSockets for candidate–recruiter chat

Updated Billing Plan (Screening + Reports)

Total Example: KES 450,000 (adjusted upward since reports are heavy)

Milestone	Deliverable	Payment %	Amount (KES)
1	Kickoff + system architecture update for reporting module	20%	90,000
2	Resume parsing + candidate profile features	20%	90,000
3	Candidate screening + scoring system	20%	90,000
4	Reporting module (core 6–7 reports + graphs/charts)	30%	135,000
5	RSS/social feed integration	10%	45,000
6	Delivery + training on reports, UAT handover	10%	45,000

